DAY-4

Hands-on Subnetting Practical (1 hour): Divide a network into subnets and assign IP addresses to devices. Task: Document the steps to create a subnet plan.

1.Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

[ANS]

Step 1: Calculate the New Subnet Mask

- Original Network: 192.168.10.0/24
- Required Subnets: 4
- Borrowed Bits: 2 (2^2 = 4 subnets)

New CIDR: /26

Subnet Mask: 255.255.255.192

Subnet Table

Subnet	Network Address	Subnet Mask	Broadcast Address
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.63
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.127
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.191
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.255

2.For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

[ANS]

Selected Subnet

- Network Address: 192.168.10.0/26
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63

Device IP Assignment

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.10
Laptop 2	192.168.10.11
Printer	192.168.10.20
Mobile Phone	192.168.10.30

Note: The network address (192.168.10.0) and broadcast address (192.168.10.63) are not assigned to any device.

3.Draw a simple diagram showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

[ANS]

Subnet Plan Diagram

```
              Main Network
            192.168.10.0/24
                  |
   ---------------------------------------------------------------
   |             |             |            |
   |             |             |            |
Subnet 1      Subnet 2      Subnet 3      Subnet 4
192.168.10.0/26   192.168.10.64/26   192.168.10.128/26   192.168.10.192/26
Hosts: 1-62     Hosts: 65-126     Hosts: 129-190     Hosts: 193-254


   |
   |
 Router (192.168.10.1)
   |
 -----------------------------------------------
 |         |         |         |
 |         |         |         |
Laptop 1    Laptop 2    Printer    Mobile
192.168.10.10 192.168.10.11 192.168.10.20 192.168.10.30
```

IP Range of Each Subnet

Subnet	Usable Host Range
Subnet 1	192.168.10.1 – 192.168.10.62
Subnet 2	192.168.10.65 – 192.168.10.126
Subnet 3	192.168.10.129 – 192.168.10.190
Subnet 4	192.168.10.193 – 192.168.10.254

4. **Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan.**

[ANS]

Step 1: Identify the Original Network

The given network is **192.168.10.0/24**. A /24 network contains 256 IP addresses.

Step 2: Determine the Number of Required Subnets

We need **4 equal subnets**.

To create 4 subnets:

- $2^2 = 4$
- Therefore, borrow 2 bits from the host portion.

Step 3: Calculate the New Subnet Mask

Original CIDR = /24

Borrowed Bits = 2

New CIDR = /26

Subnet Mask = **255.255.255.192**

Step 4: Calculate Hosts per Subnet

Remaining Host Bits:

32 - 26 = 6

Formula:

Hosts per Subnet = $2^6 - 2$

Hosts per Subnet = 64 - 2

Hosts per Subnet = **62 usable hosts**

Step 5: Calculate Subnet Ranges

Each subnet contains 64 IP addresses.

Subnet increments:

- 192.168.10.0
- 192.168.10.64
- 192.168.10.128
- 192.168.10.192

Step 6: Assign Device IP Addresses

For Subnet 1 (192.168.10.0/26):

- Router → 192.168.10.1
- Laptop 1 → 192.168.10.10
- Laptop 2 → 192.168.10.11
- Printer → 192.168.10.20
- Mobile → 192.168.10.30

Step 7: Verify the Assignment

- No device uses the network address (192.168.10.0).
- No device uses the broadcast address (192.168.10.63).
- All assigned IP addresses fall within the valid host range.

Therefore, the subnet plan is correct and all devices can communicate within the subnet.